6.2 Bisectors of Triangles

Name _____ Period ____ Date _____

Geometry 6.2 (30-minute plan) · Student sheet

I can…

- use the Circumcenter Theorem to find distances to the vertices.
- find the circumcenter of a triangle in the coordinate plane.
- use the Incenter Theorem to find distances to the sides.

Vocabulary · fill in the missing word

Concurrent — three or more lines, rays, or segments that _____ in the same point

Point of concurrency — the point where concurrent lines, rays, or segments _____

Circumcenter — the point of concurrency of the _____ bisectors of a triangle

Incenter — the point of _____ of the angle bisectors of a triangle

Part 1 · Notes and examples (fill in each note, then work the example)

Circumcenter Theorem: the circumcenter of a triangle is equidistant from the _____ of the triangle.

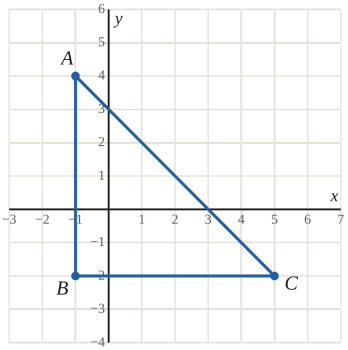
Example 1

Point P is the circumcenter of $\triangle ABC$. $PA = 3x + 2$ and $PB = 5x - 8$. Find x and PC.

Where the circumcenter is: inside an acute triangle, at the midpoint of the hypotenuse of a right triangle, and _____ an obtuse triangle.

Example 2

Find the circumcenter of the triangle with vertices A(−1, 4), B(−1, −2), and C(5, −2).



Incenter Theorem: the incenter of a triangle is equidistant from the _____ of the triangle, and it is always inside.

Which point? The same distance from three points: the circumcenter. The same distance from three lines or sides: the _____.

Example 3

Point N is the incenter of $\triangle ABC$. D, E, and F are where the perpendicular segments from N meet the sides. ND = 2x − 1 and NE = x + 4. Find x and NF.

Part 2 · Partner practice

Partner practice

The angle bisectors of $\triangle ABC$ meet at P. Point K is on $\overline{AB}$ with $\overline{PK} \perp \overline{AB}$. PA = 10 and AK = 8. Find the distance from P to $\overline{BC}$.